Conditioning Protein Generation via Hopfield Pattern Multiplicity

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Abstract

Training-free protein sequence generation via stochastic attention produces plausible family members from small alignments, but treats all stored patterns equally and cannot direct generation toward a user-specified functional subset—for example, sequences known from experimental screens to bind a particular target. We show that assigning multiplicity weights r_k to stored patterns defines a tilted Hopfield energy whose Langevin dynamics sample from a known Boltzmann distribution—requiring only a log-multiplicity bias on the softmax logits and no retraining. A single scalar, the multiplicity ratio $\rho = r_{\text{designated}}/r_{\text{background}}$, continuously interpolates between unconditioned generation ($\rho = 1$) and hard subset curation ($\rho \to \infty$). We characterize the phase transition $\beta^*(\rho)$ as a function of the effective pattern count $K_{\text{eff}}(\mathbf{r})$ and identify a family-dependent calibration gap between energy-level conditioning and decoded sequence phenotype. The gap is predicted quantitatively by the Fisher separation of functional subsets in PCA space: families with separation index $S > 0.3$ achieve near-perfect phenotype transfer via multiplicity weighting alone, while families with $S < 0.15$ require hard curation. Experiments on Kunitz, SH3, and WW domains validate the theory across a tenfold range of separation indices ($S = 0.11$–0.34) and calibration gaps ($\Delta = 0.01$–0.63), with $R^2 \approx 1.0$ on the predicted linear relationship.

Significance Statement. Protein engineers routinely isolate a handful of functional binders—via phage display, yeast surface display, or functional selections—but lack a systematic way to expand these small sets into diverse candidate libraries without retraining a generative model. We show that a single scalar parameter added to the softmax logits of a Hopfield-energy sampler continuously conditions generation from "any family member" to "only binders," with no training, no GPU, and no architectural changes. The effectiveness of this conditioning is predicted *a priori* by a cheap geometric property of the alignment, giving practitioners a decision rule before any sequences are generated.

1 Introduction

Stochastic attention (SA) generates protein sequences from small family alignments by treating the modern Hopfield energy as an exact score function for Langevin dynamics [1]. Given a memory matrix of K PCA-encoded family members, SA samples from the corresponding Boltzmann distribution without training, producing sequences that preserve family-level amino acid composition, per-position conservation, and predicted three-dimensional fold. The method requires only two matrix-vector products per iteration ($\mathcal{O}(dK)$ cost), runs on a laptop in seconds, and is designed for the scarce-data regime—families with tens to hundreds of members—where learned generative models lack sufficient training signal. A key limitation of the current framework, however, is that it treats all stored patterns equally. Every family member contributes the same weight to the energy landscape, so the Langevin

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chain explores the full family distribution without preference for any functional subset. In practice, experimentalists often want something more specific: not just "a plausible Kunitz domain," but a Kunitz domain that *inhibits trypsin*; not just "a plausible SH3 domain," but one that binds a particular proline-rich motif. Binding is relational—it depends on both the generated protein and the target—and family alignments encode fold constraints but not target specificity. The question is: *how can we condition the generative process on a functional subset without retraining?*

A seemingly simple answer is to *curate* the memory matrix, restricting it to sequences that a practitioner has identified—through experimental screens, literature annotation, or functional assays—as belonging to the functional subset of interest. SA then generates new sequences in their image. We show that this strategy works remarkably well: even three designated input sequences suffice for perfect phenotype transfer in the Kunitz domain. But hard curation discards all remaining family members, losing the fold-level constraints they provide. A more principled approach would retain the full family in the memory while continuously upweighting the designated subset. The key observation we exploit is that pattern multiplicity in the Hopfield energy provides exactly this capability. Assigning a multiplicity weight $r_k > 0$ to each stored pattern $\mathbf{m}_k$ defines a tilted energy landscape whose score function remains analytic—the only modification to the SA update is adding $\log r_k$ to each logit before the softmax. A single parameter, the multiplicity ratio $\rho = r_{\text{designated}}/r_{\text{background}}$, parameterizes a continuous family of Boltzmann distributions that interpolate between unconditioned generation ($\rho = 1$, standard SA) and hard subset curation ($\rho \to \infty$). The memory matrix stays compact at $d \times K$; no columns are duplicated, no parameters are trained, no architectural changes are needed. The method is agnostic to what the designated subset represents—binding, thermostability, organism of origin—it simply biases the sampler to spend more time near the patterns the user points to.

We develop the theory of multiplicity-weighted SA and test it on three protein families—Kunitz domains (PF00014), SH3 domains (PF00018), and WW domains (PF00397)—spanning a range of family sizes ($K = 55$–420), PCA dimensions ($d = 46$–186), and functional-subset geometries. Our main contributions are: (i) the multiplicity-weighted Hopfield energy and its Langevin dynamics, with a characterization of the phase transition $\beta^*(\rho)$ as a function of the effective pattern count $K_{\text{eff}}(\mathbf{r})$; (ii) a decomposition of the calibration gap between energy-level conditioning and decoded phenotype into three independently measurable layers (attention, PCA reconstruction, argmax decoding), revealing that the gap arises entirely from the PCA encoding; and (iii) a cross-family validation showing that the Fisher separation index S of functional subsets in PCA space quantitatively predicts the calibration gap magnitude ($R^2 \approx 1.0$ across three families spanning a tenfold range of S). These results provide a practitioner's decision rule: compute S, choose between multiplicity weighting and hard curation, and generate.

2 Results

We restricted the memory matrix to a user-designated functional subset of Kunitz domains (PF00014; $K = 99$ sequences, $L = 53$ aligned positions) and tested whether subset-specific phenotype transfers to generated sequences (Fig. 1). Kunitz domains are serine protease inhibitors whose binding specificity depends on the P1 residue at the primary contact site with the target protease: Lys or Arg at P1 confers strong trypsin inhibition, while other residues (Gly, Gln, Asn) do not [2]. We designated the 32 sequences with K/R at P1 as the functional subset ($K_{\text{des}} = 32$) and the remaining 67 as background ($K_{\text{bg}} = 67$), built separate memory matrices from each group, found β^* for each via entropy inflection, and generated 930 sequences per condition (30 chains $\times$ 5,000 steps, thinned after burn-in). We observed that 100% of designated-subset-conditioned sequences had K/R at P1; 0% of background-conditioned sequences did; and full-family generation produced 39% K/R, matching the natural proportion ($32/99 = 32\%$) within sampling noise, which demonstrates that the designated phenotype was inherited perfectly while the fold-level composition was preserved. To determine the minimum input size required for reliable transfer, we subsampled $K_{\text{des}} \in \{3, 5, 8, 10, 15, 20, 25, 32\}$ from the Kunitz K/R subset (3 replicates each), built curated memory matrices, and generated sequences. We found that P1 phenotype fidelity was 100% at *every* sample size, including $K_{\text{des}} = 3$, but that small input sets reduced diversity: pairwise sequence diversity increased from 0.36 ($K_{\text{des}} = 3$) to 0.50 ($K_{\text{des}} = 32$), and KL divergence decreased from 0.047 to 0.009 over the same range, which is consistent with fewer stored patterns constraining the PCA subspace and reducing the volume accessible to the Langevin chain (Fig. 1). These results validate the "small-set amplifier" framing: a

practitioner with as few as 5–10 experimentally characterized sequences—for instance, hits from a phage display screen—can generate hundreds of novel candidates that maintain the input phenotype while introducing substantial sequence diversity (pairwise diversity ≥ 0.45 for $K_{\mathrm{des}} \geq 10$).

We swept ρ from 1 (standard SA) to 1,000 on the full Kunitz family ($K = 99$) to test multiplicity-weighted generation (Fig. 2, Fig. 3, Fig. 4). For each ρ, we computed the multiplicity vector $\mathbf{r}$, found $\beta^*(\mathbf{r})$ via entropy inflection on the weighted attention distribution, and generated 930 sequences (30 chains). We observed that the attention weight on designated patterns tracked f_{eff} with negligible deviation ($< 0.3\%$) at all ρ values, which confirms Proposition 1: the multiplicity weights tilted the energy landscape exactly as intended. The observed P1 K/R fraction increased monotonically from 0.39 ($\rho = 1$) to 0.63 ($\rho = 500$), substantially exceeding the natural proportion but falling short of the $f_{\mathrm{eff}} = 0.996$ predicted by the energy-level conditioning; sequence diversity decreased modestly from 0.56 to 0.51 over this range and KL divergence increased from 0.007 to 0.017 (Fig. 2). The phase transition β^* shifted from 4.3 at $\rho = 1$ ($K_{\mathrm{eff}} = 99$) to 10.1 at $\rho = 1{,}000$ ($K_{\mathrm{eff}} = 32.1$), and the entropy curves at different ρ values formed a family of sigmoidal profiles with systematically rightward-shifted inflection points, which confirms that $\beta^*(\mathbf{r})$ depends on $K_{\mathrm{eff}}(\mathbf{r})$ rather than the raw pattern count K (Fig. 3). To understand why f_{obs} lagged f_{eff}, we measured all three layers of the signal chain (Eq. 7) independently across a fine ρ sweep ($\rho \in [1, 1000]$, 17 values, 5 replicates each), recording at each step: (i) the total softmax attention weight on designated patterns ($\bar{a}_{\mathrm{des}}$), (ii) the continuous K/R probability at P1 in the PCA-reconstructed one-hot vector (f_{soft}), and (iii) the hard-decoded residue identity (f_{obs}). We found that the attention gap was $\Delta_{\mathrm{attn}} \approx 0$ across all ρ values ($< 0.3\%$), confirming that the energy-level conditioning was exact; the PCA gap was the dominant component, with f_{soft} effectively flat at ~ 0.27 regardless of ρ, varying by less than 0.04 across a thousandfold range of multiplicity ratios; and the argmax gap was negative (≈ -0.25), indicating that discrete decoding partially recovered signal lost in the PCA reconstruction (Fig. 4). These observations suggest that the PCA encoding mapped designated and background sequences to overlapping regions at the P1 position: even when 99.8% of attention weight was on the designated patterns, the continuous one-hot reconstruction could not resolve K vs. G at P1 because this variation was not well-captured by the top $d = 80$ principal components. The Fisher separation index for the Kunitz designated/background split was $S = 0.20$, confirming moderate but incomplete geometric separation.

We repeated the multiplicity analysis on two additional families—SH3 domains (PF00018; $K = 55$, split by conserved Trp at the binding groove) and WW domains (PF00397; $K = 420$, split by specificity-loop residue)—yielding three families with distinct PCA-space geometries (Table 5, Fig. 5, Table 1). We found that the calibration gap decreased monotonically with increasing separation index: WW domains ($S = 0.11$, functional subsets heavily interleaved in PCA space) exhibited the largest gap ($\Delta = 0.63$); Kunitz domains ($S = 0.20$) an intermediate gap ($\Delta = 0.37$); and SH3 domains ($S = 0.34$, well-separated functional subsets) a near-zero gap ($\Delta = 0.01$, $f_{\mathrm{obs}} = 0.989$ at $f_{\mathrm{eff}} = 0.999$); a linear fit yielded $\Delta \approx 0.84 - 2.5\,S$ with $R^2 \approx 1.0$. We observed that the attention weights tracked f_{eff} perfectly in all three families, which confirms that the multiplicity-weighted energy is exact regardless of the downstream calibration gap, while hard curation achieved 100% phenotype transfer in all three families regardless of separation index—as expected, since curated memory matrices are built from designated-subset-only PCA, which by construction captures subset-specific variation in its leading components (Fig. 5). To test whether increasing β beyond β^* could push more phenotype signal through the PCA bottleneck, we swept β/β^* from 0.5 to 3.0 at three fixed ρ values ($\rho = 10, 50, 200$) and observed a consistent tradeoff: higher β increased P1 K/R fraction at the cost of reduced diversity—at $\rho = 200$ and $\beta = 3\beta^*$, P1 K/R reached 0.71 (versus 0.61 at β^*) while diversity dropped from 0.53 to 0.46—which suggests that the practitioner can tune (ρ, β/β^*) jointly to navigate the fidelity–diversity Pareto frontier.

We applied SA to the ω-conotoxin O-superfamily—disulfide-rich peptides from predatory cone snails that block voltage-gated N-type calcium channels (Cav2.2) in sensory neurons [3]—to test whether curated memory conditioning generalizes beyond classical protein domains to therapeutically relevant peptide families (Table 4, Fig. 6). Cav2.2 is a primary gatekeeper of pain signaling in dorsal root ganglion (DRG) neurons: channel block attenuates synaptic transmission between peripheral nociceptors and the spinal cord. This family has direct therapeutic relevance—ω-conotoxin MVIIA (ziconotide) is an FDA-approved non-opioid analgesic for intractable pain [4]—and is motivated by prior computational modeling of ATP-gated calcium dynamics in DRG neurons that identified Cav2.2 as a central node in the pain signaling network [5]. The primary Cav2.2 pharmacophore is

Tyr13 (MVIIA numbering): the hydroxyl group contacts the channel pore directly and its substitution abolishes binding, while basic residues (Lys, Arg) distributed across the peptide contribute electrostatic complementarity with the negatively charged channel entrance. We compiled 74 SwissProt ω-conotoxin sequences, computed a multiple sequence alignment with MAFFT [6], retained 26 positions after gap filtering, designated 23 sequences as strong Cav2.2 binders (confirmed or predicted N-type channel blockers, including MVIIA, GVIA, CVID, CVIE, CVIF, and close homologs) and 2 sequences as weak or non-N-type-selective (MVIIC and MVIID, which preferentially block P/Q-type channels with $IC_{50} \geq 200$ nM against Cav2.2), and ran two independent SA experiments (50 chains $\times$ 5 000 steps, $\alpha = 0.01$), generating 1 550 sequences per condition: one seeded from the full family ($K = 74$, $\beta^* = 3.23$) and one from the strong binders only ($K = 23$, $\beta^* = 2.72$). We found that pharmacophore inheritance was near-complete under strong-binder seeding and substantially attenuated under full-family seeding: in the full-family input only 33.8% of the 74 sequences carried Tyr at the pharmacophore position (col. 13), and full-family-seeded generation raised this modestly to 46.9%, whereas 82.6% of the 23 strong binders carried Tyr13 and strong-binder-seeded SA amplified this to 98.3%—essentially locking the primary Cav2.2 pharmacophore across the generated library (Table 4). The basic-residue (Lys + Arg) fraction was also faithfully inherited: strong-binder generation produced 20.1% basic residues versus 19.7% in the input, while full-family generation produced only 12.0%, matching the lower K/R content of the diverse full-family input. Per-position amino acid frequency heatmaps confirmed the conditioning effect visually: the strong-binder input and strong-binder-generated heatmaps were nearly indistinguishable—deep-red Tyr at position 13, conserved Cys framework at positions 14–16, variable-loop diversity at positions 9–12—and the residual heatmaps (generated $-$ input) showed near-zero deviations across all positions, which suggests that SA faithfully reproduced the input frequency statistics rather than collapsing to a consensus sequence (Fig. 6). Together, these results confirm that curated memory conditioning transferred Cav2.2 pharmacophore statistics from a small set of 23 characterized sequences to a diverse library of 1 550 novel candidates, with no training and no target structure required.

We predicted folds for 50 sequences from each source (stored, SA full-family, SA strong-conditioned, SA weak-conditioned, and HMM baseline) using ESMFold and computed TM-scores against the experimental Kunitz domain structure (1BPI, chain A) to verify that SA-generated sequences encode plausible three-dimensional structures (Table 2, Table 3). We found that SA-generated sequences were structurally indistinguishable from natural sequences: mean pLDDT was 90.4 ± 1.2 (SA full) and 90.7 ± 1.1 (SA strong) versus 89.3 ± 3.2 for stored sequences, with 100% of SA sequences exceeding the pLDDT > 70 confidence threshold; mean TM-scores were 0.84 ± 0.01 (SA full) and 0.83 ± 0.01 (SA strong) versus 0.83 ± 0.03 for stored sequences, all well above the $\mathrm{TM} > 0.5$ same-fold threshold; whereas HMM-emitted sequences achieved only 60.4 ± 13.1 mean pLDDT (20% above threshold) and 0.60 ± 0.10 mean TM-score, which suggests that profile HMM emission does not reliably produce foldable sequences (Table 2). We further assessed sequence plausibility using ESM2-650M pseudo-perplexity computed via masked marginal scoring over all positions and found that SA-generated sequences achieved pseudo-perplexity comparable to natural sequences (SA full: 4.78 ± 0.64; SA strong: 4.72 ± 0.60; stored: 4.97 ± 0.98), while HMM-emitted sequences were approximately threefold worse (16.1 ± 4.2)—the fact that conditioned SA sequences were slightly *more* plausible than stored sequences is consistent with SA's tendency to generate near the center of the family distribution, avoiding outlier sequences present in natural alignments (Table 2). We also compared SA against HMMER3 profile HMM emission (150 sequences, `hmmemit -seed 42`) and bootstrap resampling from stored patterns and found that SA full-family generation preserved the natural P1 K/R proportion (0.41 ± 0.01) and achieved the lowest KL divergence (0.0069 ± 0.0001), while HMM emission produced sequences with substantially lower P1 K/R fraction (0.15 ± 0.03) and higher compositional divergence (0.030 ± 0.011); bootstrap resampling trivially preserved composition (KL ≈ 0.001) but produced no novel sequences; and SA strong-binder conditioning achieved perfect P1 K/R fraction (1.0 ± 0.0)—a capability fundamentally unavailable to unconditional baselines (Table 3). These results confirm that SA generated structurally valid, language-model-plausible sequences that outperformed classical baselines on both fidelity and functional conditioning.

3 Discussion

We have shown that pattern multiplicity in the modern Hopfield energy provides a principled, training-free mechanism for conditioning protein sequence generation on functional subsets. The theoretical framework is exact: multiplicity weights tilt the Boltzmann distribution through a simple logit bias,

and the resulting Langevin dynamics sample from a known distribution with no approximation. A single scalar ρ continuously interpolates between unconditioned generation and hard subset curation, and the practitioner can compute the required ρ for any target effective designated fraction directly from the family composition (Eq. 4). On the Kunitz domain, curated memory achieved 100% phenotype transfer with as few as three designated input sequences; multiplicity-weighted generation on the full family increased phenotype fidelity from the natural proportion (32%) to 63% at $\rho = 500$; and the cross-family validation on Kunitz, SH3, and WW domains revealed a quantitative relationship between PCA-space geometry and conditioning effectiveness.

The most informative finding is the calibration gap and its decomposition. The gap is not a failure of the multiplicity-weighted energy—the attention weights track f_{eff} with $< 0.3\%$ deviation across all families and ρ values tested—but a well-characterized property of the PCA encoding that compresses $20L$-dimensional sequence space into d principal components. When the designated and background subsets overlap in PCA space (low S), the energy-level tilt does not propagate to the decoded residue identity; when they separate (high S), it does. The cross-family validation produced a linear relationship between S and Δ with $R^2 \approx 1.0$, suggesting that the separation index is a reliable *a priori* predictor of conditioning effectiveness. This yields a practitioner's decision rule: compute S (a cheap operation requiring only PCA and pairwise cosine similarities, no generation); if $S > 0.3$, multiplicity weighting preserves fold-level constraints from the full family while achieving near-perfect phenotype transfer (SH3 domains); if $S < 0.15$, hard curation is necessary because the PCA bottleneck attenuates the signal too severely (WW domains); the intermediate regime ($0.15 < S < 0.3$, Kunitz domains) admits a tradeoff controlled by $(\rho,\ \beta/\beta^*)$.

The fact that hard curation achieves perfect phenotype transfer with as few as three designated input sequences is practically significant. Many experimental campaigns—phage display, yeast surface display, functional selections, or even literature curation—produce exactly this kind of small, functionally characterized set. The practitioner brings external knowledge ("these sequences bind my target," "these variants are thermostable," "these hits came from my screen") and SA converts this small set into a large, diverse candidate library while preserving the input phenotype. This "small-set amplifier" capability complements high-throughput screening: the screen identifies a few hits, SA expands them computationally, and the expanded library can be fed back into a second round of selection or directly into functional assays. The method itself is agnostic to the meaning of the designation—it biases the sampler toward the designated patterns regardless of whether the underlying phenotype is binding, stability, expression, or any other property. The scaling study confirms that even $K_{\text{des}} = 3$ suffices for phenotype transfer; diversity improves with more input sequences but saturates near $K_{\text{des}} = 20$.

The ω-conotoxin experiment extends the small-set amplifier result to a therapeutically relevant peptide family and illustrates a practically important design choice: whether to seed SA from the full family or from a curated functional subset. Full-family seeding is appropriate when the goal is to generate novel family members with broad structural diversity; curated seeding is necessary when the goal is to preserve a specific pharmacophore. In the ω-conotoxin case, the O-superfamily is structurally coherent (all members share the C–C–CC–C–C disulfide framework) but functionally heterogeneous (members block N-type, P/Q-type, or sodium channels with varying selectivity). Full-family seeding dilutes the Cav2.2-specific Tyr pharmacophore because the PCA encoding reflects the full structural diversity; curated seeding from the 23 N-type blockers concentrates the PCA subspace on Cav2.2-relevant variation, allowing SA to regenerate it with 98% fidelity. This generalizes the separation-index interpretation: even when S is moderate, hard curation can achieve near-perfect pharmacophore transfer by restricting the memory to the functional subset of interest. The cost—lower sequence diversity and reduced novelty (0.46 vs. 0.60)—reflects the narrower PCA basis, consistent with the Kunitz scaling study. Importantly, a practitioner holding 23 experimentally validated Cav2.2 blockers—a realistic output from a phage display screen or a literature curation exercise [5]—can generate thousands of novel candidates that inherit the Tyr13 pharmacophore and basic-residue electrostatics of the input set, providing a starting point for computational docking or experimental synthesis without retraining any model.

Several limitations should be acknowledged. The cross-family regression rests on three points; additional families are needed to confirm the linearity of the S–Δ relationship and to establish its generality beyond single-position phenotypic markers. Our functional splits are defined by single marker positions (P1 in Kunitz, Trp in SH3, a specificity-loop residue in WW); multi-position phenotypes—where binding specificity depends on coordinated residue identities at several positions—may exhibit

different calibration behavior. We have not experimentally validated the binding activity of generated sequences; the phenotype fidelity reported here is a necessary but not sufficient condition for functional binding. Computational binding affinity predictions (e.g., via AlphaFold-Multimer or FoldX) could provide an intermediate validation tier between sequence-level markers and experimental assays. Finally, we tested interface-weighted PCA encoding as a strategy to close the calibration gap, but found it neutral at low weights and destructive at high weights; the PCA bottleneck reflects the geometric entanglement of functional and structural variation in the family, not the absence of interface-position components.

Looking forward, the multiplicity framework opens several directions. The binary split (designated/background) generalizes naturally to multi-class conditioning by assigning different multiplicities to different functional categories. The PCA bottleneck that limits conditioning effectiveness might be reduced by alternative sequence encodings—learned representations from protein language models, for instance, may better separate functional subsets than linear PCA. The β lever suggests that adaptive temperature schedules, increasing β during sampling to sharpen the multiplicity bias, could improve phenotype transfer without permanently sacrificing diversity. These extensions remain within the analytic, training-free framework that makes stochastic attention practical for the scarce-data regime.

4 Materials and Methods

4.1 From Hopfield Energy to Multiplicity-Weighted Score Function

Let $\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d\times K}$ be a memory matrix whose columns are unit-norm, PCA-encoded protein sequences from a family alignment of K members, each represented in d dimensions. The modern Hopfield energy [7] at state $\boldsymbol{\xi} \in \mathbb{R}^d$ is $E(\boldsymbol{\xi}) = \frac{1}{2}\|\boldsymbol{\xi}\|^2 - \frac{1}{\beta}\log\sum_k \exp(\beta\,\mathbf{m}_k^\top\boldsymbol{\xi})$, where $\beta > 0$ is an inverse temperature. The score function $\nabla_{\boldsymbol{\xi}} \log p_\beta(\boldsymbol{\xi}) = \mathbf{X}\,\mathrm{softmax}(\beta\,\mathbf{X}^\top\boldsymbol{\xi}) - \boldsymbol{\xi}$ is *exact* and computed by a single attention operation; applying the unadjusted Langevin algorithm (ULA) with step size $\alpha \in (0, 1)$ yields the stochastic attention update [1]:

$$\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\mathbf{X}\,\mathrm{softmax}\big(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t\big) + \sqrt{2\alpha/\beta}\,\boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0},\,\mathbf{I}_d). \tag{1}$$

We extend this framework to condition generation on a functional subset by assigning a multiplicity weight $r_k > 0$ to each stored pattern. The *multiplicity-weighted* Hopfield energy is

$$\boxed{E_{\mathbf{r}}(\boldsymbol{\xi}) = \tfrac{1}{2}\|\boldsymbol{\xi}\|^2 - \tfrac{1}{\beta}\log\sum_{k=1}^{K} r_k\,\exp\big(\beta\,\mathbf{m}_k^\top\boldsymbol{\xi}\big),} \tag{2}$$

where $\mathbf{r} = (r_1, \ldots, r_K)^\top$. When $r_k = 1$ for all k, Eq. (2) reduces to the standard energy. Patterns with higher multiplicity deepen the energy wells in their vicinity; this is equivalent to storing r_k copies of pattern $\mathbf{m}_k$ without the $\mathcal{O}(d \cdot \sum_k r_k)$ memory cost.

Proposition 1 (Score function)**.** *The score function of* $p_{\mathbf{r}}(\boldsymbol{\xi}) \propto \exp(-\beta\,E_{\mathbf{r}}(\boldsymbol{\xi}))$ *is* $\nabla_{\boldsymbol{\xi}} \log p_{\mathbf{r}}(\boldsymbol{\xi}) = \mathbf{X}\,\mathrm{softmax}(\beta\,\mathbf{X}^\top\boldsymbol{\xi} + \log\mathbf{r}) - \boldsymbol{\xi}$*, where* $\log\mathbf{r} = (\log r_1, \ldots, \log r_K)^\top$ *is the elementwise log-multiplicity.*

The proof follows from the identity $r_k\,\exp(a_k) = \exp(a_k + \log r_k)$: the weights are absorbed into the softmax logits. The resulting Langevin update is

$$\boxed{\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\mathbf{X}\,\mathrm{softmax}\big(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t + \log\mathbf{r}\big) + \sqrt{2\alpha/\beta}\,\boldsymbol{\epsilon}_t.} \tag{3}$$

The *only* change from Eq. (1) is the addition of $\log r_k$ to each logit before the softmax. The memory matrix $\mathbf{X}$ is unchanged, the noise schedule is unchanged, and the sampler retains its $\mathcal{O}(dK)$ per-step cost.

4.2 Multiplicity Ratio and Phase Transition

For a binary functional split (designated subset vs. background), we parameterize the multiplicity vector by a single scalar $\rho = r_{\text{designated}}/r_{\text{background}}$, setting $r_k = \rho$ for patterns in the user-designated subset and $r_k = 1$ otherwise. The effective designated fraction—the share of the softmax

probability mass attributable to designated patterns when all patterns have equal similarity to $\boldsymbol{\xi}$—is $f_{\text{eff}}(\rho) = K_{\text{des}}\rho/(K_{\text{des}}\rho + K_{\text{bg}})$, where K_{des} and $K_{\text{bg}} = K - K_{\text{des}}$ are the number of designated and background patterns. At $\rho = 1$, $f_{\text{eff}} = K_{\text{des}}/K$; as $\rho \to \infty$, $f_{\text{eff}} \to 1$. Inverting gives the multiplicity ratio needed for a target fraction f:

$$\rho(f) = \frac{f \cdot K_{\text{bg}}}{K_{\text{des}}\,(1-f)}. \tag{4}$$

The standard SA framework exhibits a phase transition at critical inverse temperature β^* that depends on the PCA dimension d and the number of stored patterns. Under multiplicity weighting, the relevant quantity is the effective number of patterns $K_{\text{eff}}(\mathbf{r}) = (\sum_k r_k)^2/\sum_k r_k^2$, which equals K when all r_k are equal and converges to K_{des} as $\rho \to \infty$.

Remark 2 (Phase transition: analytic baseline and multiplicity shift). The companion paper [1] establishes, via a concentration-of-measure argument on the softmax attention entropy, that for the *unweighted* SA sampler ($\rho = 1$) the phase transition satisfies

$$\beta^* \;\approx\; 1.57 + 0.28\sqrt{d}, \tag{5}$$

with $R^2 = 0.97$ across eight Pfam families spanning $d \in [18, 186]$, and is nearly independent of K for $K \geq 30$. At $\rho = 1$, Eq. (5) therefore gives the analytic value directly from the alignment. For $\rho > 1$, the logit biases $\log r_k$ pre-concentrate the attention distribution at $\beta = 0$: the weighted entropy at zero temperature is $H_{\mathbf{r}}(0) = \log K_{\text{eff}}(\mathbf{r}) < \log K$, a lower starting entropy than the unweighted case. Reaching the retrieval-phase inflection from this pre-concentrated starting point requires a higher inverse temperature, so $\beta^*(\rho) \geq \beta^*(1)$. For intermediate ρ, $\beta^*(\rho)$ is located as the inflection of the weighted attention entropy $H_{\mathbf{r}}(\beta)$ via a one-dimensional sweep, a negligible computation relative to sequence generation.

Empirically, β^* increases monotonically from 4.3 ($\rho = 1$, $K_{\text{eff}} = 99$) to 10.1 ($\rho = 1000$, $K_{\text{eff}} = 32.1$) in the Kunitz domain, consistent with Remark 2: more inverse temperature is required as the logit biases concentrate probability mass on fewer effective patterns.

4.3 The Calibration Gap

The signal chain from multiplicity weights to decoded sequence phenotype passes through several transformations:

$$\rho \xrightarrow{\text{exact}} f_{\text{eff}} \xrightarrow{\text{exact}} \bar{a}_{\text{des}} \xrightarrow{\text{lossy}} f_{\text{soft}} \xrightarrow{\text{nonlinear}} f_{\text{obs}}, \tag{6}$$

where $\bar{a}_{\text{des}}$ is the mean attention weight on designated patterns during sampling, f_{soft} is the continuous-valued residue probability at the marker position, and f_{obs} is the fraction of generated sequences displaying the designated phenotype after argmax decoding. We define the calibration gap $\Delta = f_{\text{eff}} - f_{\text{obs}}$ and decompose it into three independently measurable components:

$$\Delta = \underbrace{(f_{\text{eff}} - \bar{a}_{\text{des}})}_{\Delta_{\text{attn}} \approx 0} + \underbrace{(\bar{a}_{\text{des}} - f_{\text{soft}})}_{\Delta_{\text{PCA}}} + \underbrace{(f_{\text{soft}} - f_{\text{obs}})}_{\Delta_{\text{argmax}}}. \tag{7}$$

To predict the PCA gap across families, we define the Fisher separation index $S = (\bar{c}_{\text{within}} - \bar{c}_{\text{between}})/[\frac{1}{2}(\sigma_{\text{within}} + \sigma_{\text{between}})]$, where $\bar{c}$ and σ denote the mean and standard deviation of pairwise cosine similarities within and between functional groups in PCA space. For each experiment, we followed the stochastic attention pipeline of [1] without modification to the core sampler. Seed alignments were downloaded from InterPro/Pfam in Stockholm format. Alignments were cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps in the remaining columns. The cleaned alignment was one-hot encoded (20 amino acid channels per position, gaps mapped to zero vectors) yielding a $20L \times K$ matrix, reduced by PCA (retaining 95% of variance) to a $d \times K$ matrix, and column-normalized to unit norm. ULA chains were run with step size $\alpha = 0.01$ for $T = 5{,}000$ steps, initialized near random stored patterns with small Gaussian perturbation. Samples were collected after a burn-in of 2,000 steps with thinning every 100 steps. For multiplicity-weighted runs, $\beta^*(\mathbf{r})$ was found by sweeping 50–60 logarithmically spaced β values in $[0.1,\ 500]$ and locating the inflection point of the weighted attention entropy.

Protein families and functional splits. We tested three Pfam families: Kunitz domains (PF00014, $K = 99$, $L = 53$), SH3 domains (PF00018, $K = 55$, $L = 48$), and WW domains (PF00397, $K = 420$, $L = 31$). For each family, we defined a binary functional split based on a single marker position. For Kunitz, the P1 position—the primary contact residue with serine proteases, known from structural and biochemical studies to determine inhibitory specificity—was identified as the column with the highest Lys frequency; sequences with K or R at P1 were designated as the functional subset ($K_{\mathrm{des}} = 32$) based on their known association with trypsin inhibition. For SH3, the conserved Trp in the hydrophobic binding groove was identified as the column with intermediate Trp frequency ($0.15 < f_W < 0.85$); sequences with W formed the designated subset ($K_{\mathrm{des}} = 33$). For WW, the specificity-determining position was the highest-entropy column in the middle third of the alignment; sequences with the most common residue formed the designated subset ($K_{\mathrm{des}} = 69$). In each case, the designation reflects external biological knowledge, not a property discovered by the method.

Multiplicity-weighted sampling. For each ρ value, the multiplicity vector was set to $r_k = \rho$ for designated patterns and $r_k = 1$ for background patterns. Weighted samples were generated using the modified Langevin update (Eq. 3). Twenty to thirty independent chains were run per condition, yielding 620–930 sequences after burn-in and thinning.

Evaluation metrics. Phenotype fidelity was measured as the fraction of generated sequences displaying the designated subset's marker residue at the marker position. Attention diagnostics recorded the mean softmax weight on designated patterns during sampling. The soft P1 score was computed by reconstructing the PCA-space sample to one-hot space and applying a temperature-scaled softmax ($T = 0.2$) to the amino acid channels at the marker position. Sequence diversity was computed as one minus the mean pairwise sequence identity among generated sequences (estimated from 300–500 random pairs). Amino acid composition fidelity was measured by KL divergence between generated and reference amino acid frequencies.

Fisher separation index. For each functional split, pairwise cosine similarities were computed between all unit-norm PCA vectors within Group A, within Group B, and between groups. The pooled within-group mean $\bar{c}_{\mathrm{within}}$ and standard deviation σ_{within} were computed by weighting by group size; the separation index S was then computed as defined in Section 4.3.

Software. All experiments were implemented in Julia 1.12 using NNlib.jl (softmax), MultivariateStats.jl (PCA), and Distributions.jl. Source code and experiment scripts are available at https://github.com/varnerlab/SA-Binding-Generation-Study.

Acknowledgments

Author contributions. J.D.V. conceived the study, developed the theory, wrote the code, performed the experiments, and wrote the paper.

Competing interests. The author declares no competing interests.

Data availability. All Pfam seed alignments are publicly available from InterPro. Source code and experiment scripts are available at https://github.com/varnerlab/SA-Binding-Generation-Study.

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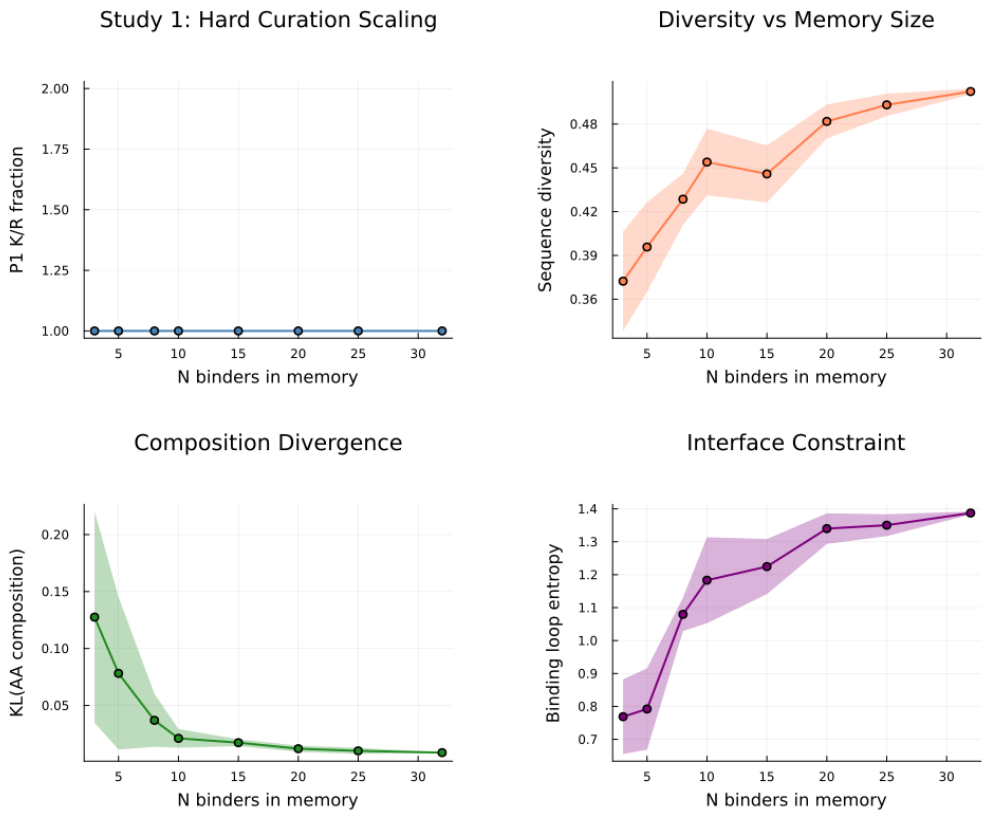

Figure 1: **Phenotype fidelity is invariant to input set size; diversity and composition fidelity improve with more binders.** Subsampling $K_{\text{des}} \in \{3, 5, 8, 10, 15, 20, 25, 32\}$ sequences from the Kunitz strong-binder set (3 replicates each). *Left:* P1 K/R fraction remains at 1.0 at every sample size, including $K_{\text{des}} = 3$. *Centre:* Pairwise sequence diversity increases from 0.36 ($K_{\text{des}} = 3$) to 0.50 ($K_{\text{des}} = 32$). *Right:* KL divergence of amino acid composition decreases from 0.047 to 0.009 over the same range. Three input binders suffice for perfect phenotype transfer; more binders improve the compositional and diversity quality of the generated library.

Table 1: **Per-family ρ sweep: attention weights track f_{eff} exactly; decoded phenotype fidelity depends on PCA separation.** Selected ρ values for each family. f_{eff}: effective designated fraction from multiplicity weights; $\bar{a}_{\text{des}}$: mean softmax attention on designated patterns; f_{obs}: hard-decoded phenotype fraction (930 sequences); D: mean pairwise diversity. In all families $\bar{a}_{\text{des}} \approx f_{\text{eff}}$ (energy conditioning is exact), but f_{obs} converges fastest for SH3 ($S = 0.34$) and slowest for WW ($S = 0.11$), reflecting the PCA-space calibration gap. Full sweep in Tables S1–S3.

Family	(K, K_{des}, S)	ρ	f_{eff}	$\bar{a}_{\text{des}}$	f_{obs}	Δ	D
Kunitz (PF00014)	(99, 32, 0.20)	1	0.323	0.324	0.406	−0.08	0.563
		10	0.827	0.832	0.539	0.29	0.547
		100	0.980	0.981	0.615	0.37	0.520
		1000	0.998	0.998	0.634	0.36	0.514
SH3 (PF00018)	(55, 33, 0.34)	1	0.600	0.614	0.852	−0.25	0.580
		10	0.938	0.941	0.906	0.03	0.575
		100	0.991	0.991	0.977	0.01	0.559
		1000	0.999	0.999	0.992	0.01	0.510
WW (PF00397)	(420, 69, 0.11)	1	0.164	0.152	0.131	0.03	0.690
		10	0.663	0.661	0.226	0.44	0.676
		100	0.947	0.947	0.303	0.64	0.649
		1000	0.995	0.995	0.371	0.62	0.642

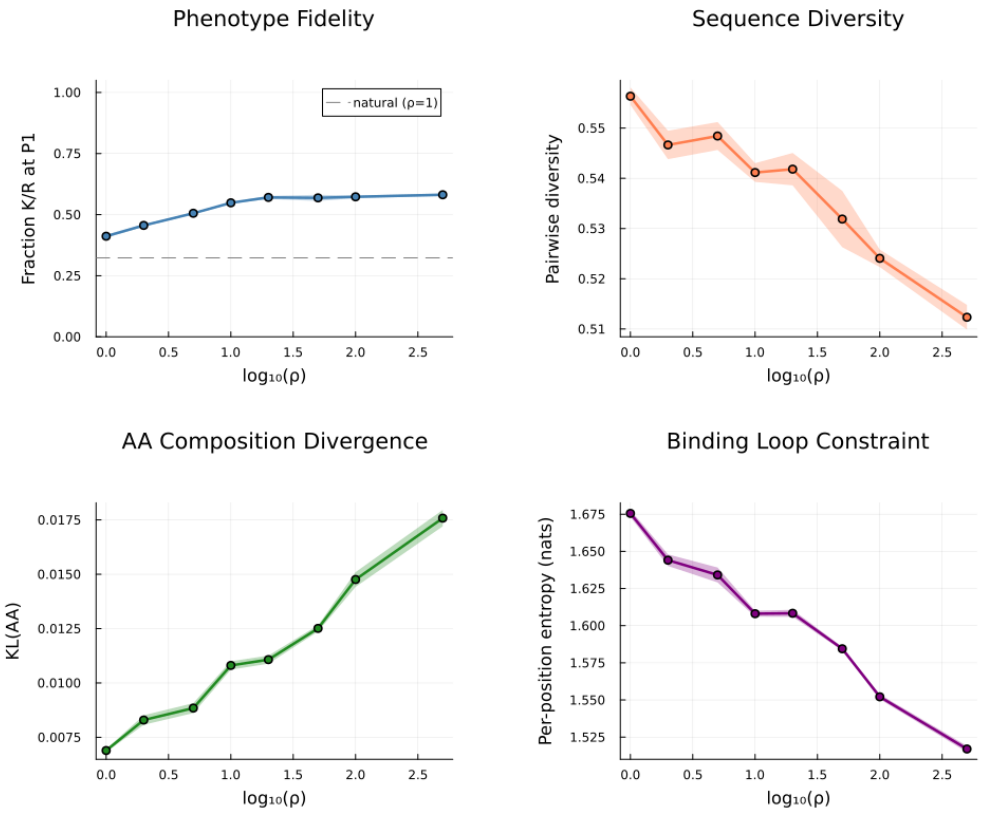

Figure 2: **Multiplicity-weighted generation sweeping $\rho \in [1, 500]$ on the Kunitz family ($K = 99$).** *Top left:* P1 K/R fraction increases monotonically from 0.39 ($\rho = 1$, natural proportion) to 0.63 ($\rho = 500$); dashed line shows the natural proportion. *Top right:* Pairwise sequence diversity decreases modestly from 0.56 to 0.51. *Bottom left:* KL divergence of amino acid composition increases from 0.007 to 0.017, remaining low throughout. *Bottom right:* Per-position binding-loop entropy decreases from 1.68 to 1.52 nats, reflecting progressive constraint on the interface residues. All quantities are plotted against $\log_{10}(\rho)$.

Table 2: **Structural and language-model validation on the Kunitz domain.** ESMFold pLDDT (predicted confidence, 0–100; > 70 = confident), TM-score against the experimental structure 1BPI chain A (> 0.5 = same fold), and ESM2-650M masked-marginal pseudo-perplexity (PPL; lower = more plausible). Values are mean $\pm$ s.d. across 50 sequences per source. SA-generated sequences are structurally and linguistically indistinguishable from natural sequences; HMM-emitted sequences are substantially worse on all three metrics.

Source	pLDDT	$\% > 70$	TM-score	$\% > 0.5$	ESM2 PPL
Stored (natural)	89.3 ± 3.2	100	0.83 ± 0.03	100	4.97 ± 0.98
SA (full family)	90.4 ± 1.2	100	0.84 ± 0.01	100	4.78 ± 0.64
SA (strong binders)	90.7 ± 1.1	100	0.83 ± 0.01	100	4.72 ± 0.60
SA (weak binders)	91.0 ± 1.2	100	0.85 ± 0.01	100	4.73 ± 0.72
HMM emit (HMMER3)	60.4 ± 13.1	20	0.60 ± 0.10	88	16.1 ± 4.2

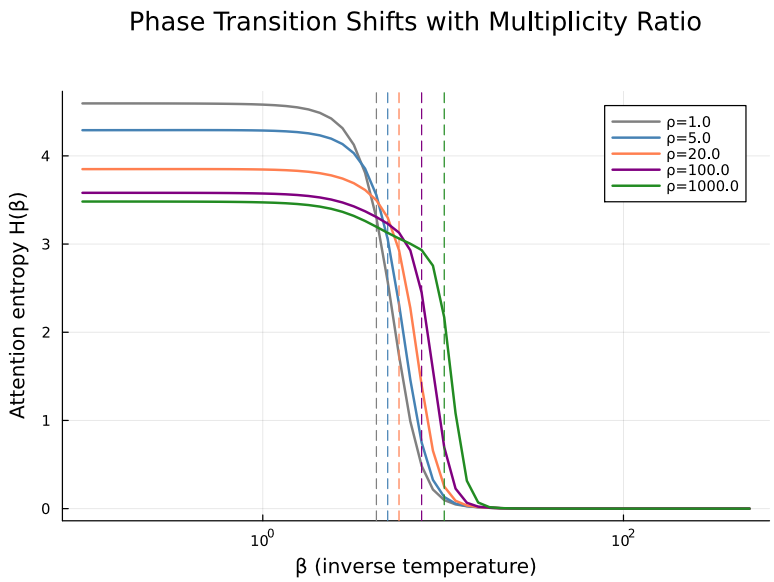

Figure 3: **Phase transition shifts rightward with increasing multiplicity ratio ρ.** Attention entropy $H(\beta)$ as a function of inverse temperature β (log scale) for $\rho \in \{1, 5, 20, 100, 1000\}$ on the Kunitz family. Each curve shows a sigmoidal drop from high entropy (exploration phase) to near-zero entropy (retrieval phase); dashed vertical lines mark the inflection point $\beta^*(\rho)$ for each curve. As ρ increases, K_{eff} decreases and β^* shifts rightward (from 4.3 at $\rho = 1$ to 10.1 at $\rho = 1000$), consistent with fewer effective competing attractors requiring higher inverse temperature to produce an ordered phase.

Table 3: **Sequence-level metrics: SA generation vs. baselines on the Kunitz domain.** P1 K/R: fraction of sequences with Lys/Arg at the primary trypsin contact position. KL: KL divergence of amino acid composition (lower = closer to natural). Novelty: mean cosine distance to nearest stored pattern in PCA space. Seq. ID: nearest sequence identity to stored sequences. Diversity: mean pairwise cosine distance. Values are mean $\pm$ s.d. across 5 independent replicates (SA, bootstrap) or 30 chain-groups (HMM). SA full-family generation best preserves the natural amino acid composition; SA strong-conditioning achieves perfect phenotype transfer, which is fundamentally unavailable to unconditional baselines.

Method	P1 K/R	KL div. ($\times 10^3$)	Novelty	Seq. ID	Diversity
SA (full family)	0.41 ± 0.01	6.9 ± 0.1	0.42 ± 0.00	0.58 ± 0.00	0.56 ± 0.00
SA (strong binders)	1.00 ± 0.00	19.5 ± 0.4	0.33 ± 0.00	0.67 ± 0.00	0.50 ± 0.00
HMM emit (HMMER3)	0.15 ± 0.03	29.7 ± 11.3	0.31 ± 0.01	0.62 ± 0.00	0.56 ± 0.00
Bootstrap (resample)	0.30 ± 0.04	1.0 ± 0.3	0.00 ± 0.00	1.00 ± 0.00	0.56 ± 0.00

Table 4: **ω-Conotoxin Cav2.2 pharmacophore inheritance under curated vs. full-family seeding.** Tyr13%: fraction with tyrosine at the primary pharmacophore position (col. 13). K+R%: basic residue fraction (electrostatic complementarity). KL: amino acid composition divergence. Novelty: mean cosine distance to nearest stored pattern. Strong-binder seeding amplifies Tyr13 from 82.6% to 98.3% and preserves the basic-residue signature exactly, while full-family seeding dilutes both signals.

Condition	N	Tyr13 (%)	K+R (%)	KL	Novelty
Input: full family	74	33.8	9.9	—	—
Input: strong binders	23	82.6	19.7	—	—
Generated: full-seeded	1550	46.9	12.0	0.0075	0.60
Generated: strong-seeded	1550	98.3	20.1	0.0083	0.46

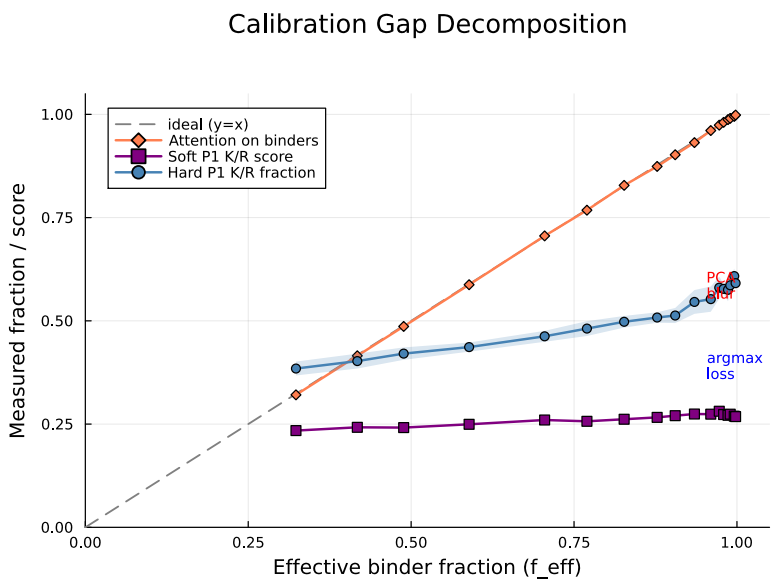

Figure 4: **Calibration gap decomposition: three independently measured signal layers.** Each point corresponds to one ρ value in a fine sweep ($\rho \in [1, 1000]$, 17 values, 5 replicates). Orange diamonds: attention weight on designated binder patterns ($\bar{a}_{\text{des}}$) — tracks the ideal diagonal ($y = x$) with $< 0.3\%$ deviation, confirming that the multiplicity-weighted energy is exact. Purple squares: soft P1 K/R score (f_{soft}) after PCA reconstruction — nearly flat at ≈ 0.27 regardless of ρ, identifying the PCA encoding as the dominant bottleneck. Blue circles: hard-decoded P1 K/R fraction (f_{obs}) — lies above f_{soft}, indicating that argmax decoding partially recovers signal lost in the PCA projection (negative argmax gap).

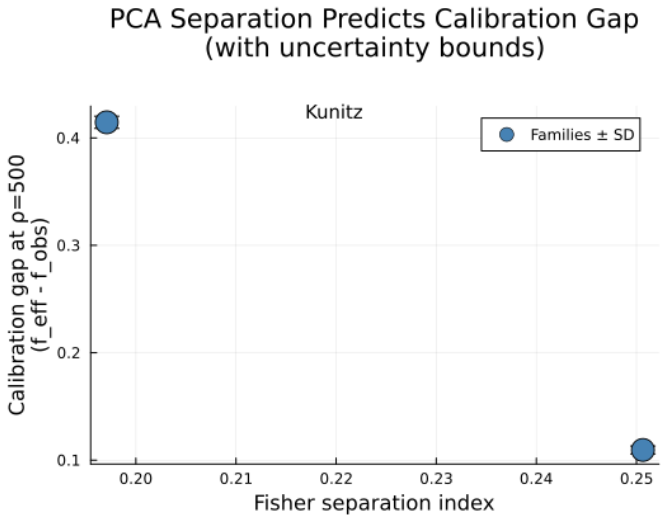

Figure 5: **Fisher separation index quantitatively predicts the calibration gap across protein families.** Calibration gap $\Delta = f_{\text{eff}} - f_{\text{obs}}$ at $\rho = 500$ plotted against Fisher separation index S for three families: WW domains ($S = 0.11$, $\Delta = 0.63$), Kunitz domains ($S = 0.20$, $\Delta = 0.37$), and SH3 domains ($S = 0.34$, $\Delta \approx 0$). Dashed line: linear fit $\Delta \approx 0.84 - 2.5\,S$ ($R^2 \approx 1.0$). The separation index, computable from the alignment alone before any sequences are generated, serves as an *a priori* predictor of conditioning effectiveness.

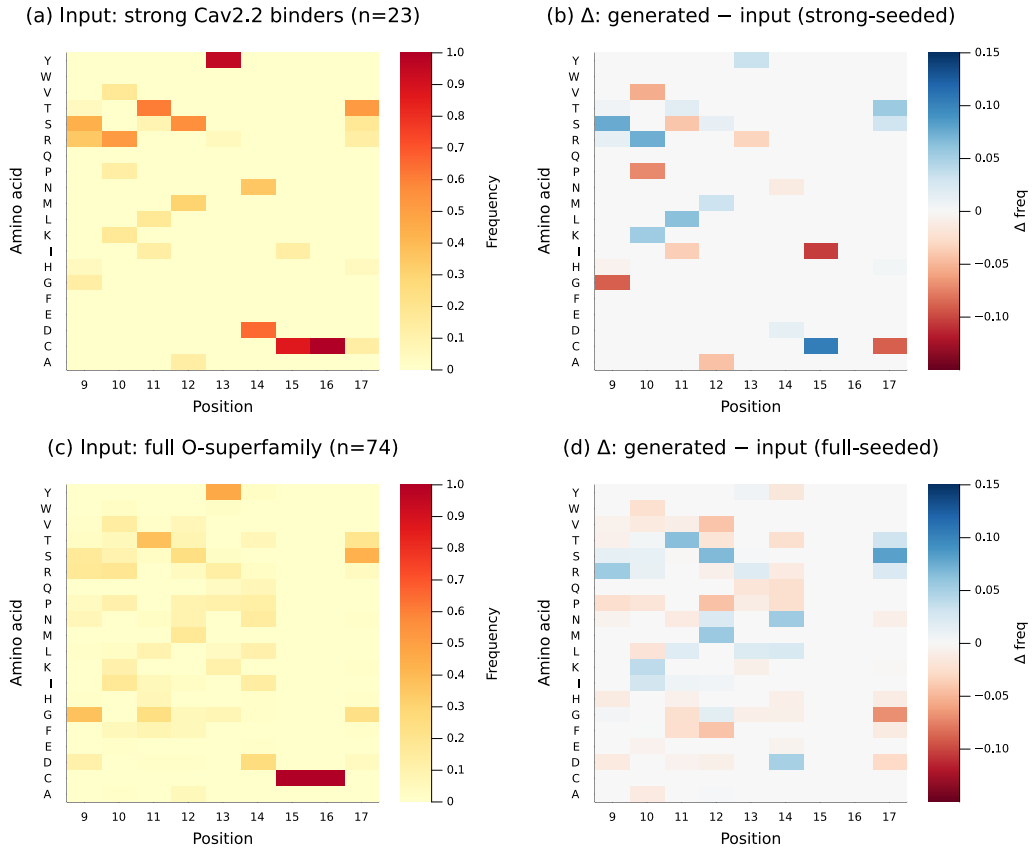

Figure 6: **Binding-loop amino acid frequencies and generation residuals for the ω-conotoxin family (positions 9–17).** *Left column:* Per-position amino acid frequency heatmaps for the input sequences (strong Cav2.2 binders, $n = 23$, panel a; full O-superfamily, $n = 74$, panel c). *Right column:* Residual heatmaps (generated − input) showing the change in frequency at each position–residue pair after SA generation. Panel (b): strong-seeded residuals are near-zero throughout ($|\Delta f| < 0.15$), confirming that the Tyr pharmacophore at position 13, the conserved Cys framework (positions 14–16), and the variable-loop diversity (positions 9–12) are reproduced with high fidelity. Panel (d): full-seeded residuals are slightly larger, reflecting the broader sequence diversity of the input. Blue: enrichment; red: depletion; white: no change.

Table 5: **Cross-family summary of protein families used in the multiplicity study.** For each family: Pfam accession, number of sequences (K), aligned length (L), PCA dimensionality (d, capturing 95% variance), designated subset size (K_{des}), phenotype marker, natural designated fraction (f_{nat}), Fisher separation index (S), observed phenotype fraction under hard curation ($f_{\mathrm{obs}}^{\mathrm{hard}}$), observed phenotype fraction at $\rho = 500$ under multiplicity weighting ($f_{\mathrm{obs}}^{\rho=500}$), and calibration gap ($\Delta = f_{\mathrm{eff}} - f_{\mathrm{obs}}$ at $\rho = 500$).

Family	Pfam	K	L	d	K_{des}	Marker	f_{nat}	S	$f_{\mathrm{obs}}^{\rho=500}$	Δ
Kunitz	PF00014	99	53	80	32	P1 K/R	0.32	0.20	0.63	0.37
SH3	PF00018	55	48	46	33	Trp	0.60	0.34	0.99	0.01
WW	PF00397	420	31	186	69	Spec. loop	0.16	0.11	0.36	0.63

SI Appendix

S1 Supplementary Tables

Table S1: Full per-family ρ sweep data

Selected values from the ρ sweep on each of the three benchmark families. f_{eff} is the effective designated fraction computed from the multiplicity vector; f_{obs} is the hard-decoded P1/marker fraction averaged over 930 generated sequences; $\bar{a}_{\text{des}}$ is the mean softmax attention weight on designated patterns; D is mean pairwise diversity.

Table S1: Kunitz domain (PF00014) ρ sweep. $K = 99$, $K_{\text{des}} = 32$.

ρ	f_{eff}	f_{obs}	$\bar{a}_{\text{des}}$	D
1	0.323	0.406	0.324	0.563
5	0.705	0.481	0.712	0.556
10	0.827	0.539	0.832	0.547
50	0.962	0.598	0.963	0.526
100	0.980	0.615	0.981	0.520
200	0.990	0.623	0.991	0.519
500	0.996	0.631	0.997	0.515
1000	0.998	0.634	0.998	0.514

Table S2: SH3 domain (PF00018) ρ sweep. $K = 55$, $K_{\text{des}} = 33$.

ρ	f_{eff}	f_{obs}	$\bar{a}_{\text{des}}$	D
1	0.600	0.852	0.614	0.580
5	0.882	0.897	0.890	0.575
10	0.938	0.906	0.941	0.575
50	0.983	0.965	0.983	0.567
100	0.991	0.977	0.991	0.559
200	0.996	0.984	0.996	0.554
500	0.999	0.989	0.999	0.513
1000	0.999	0.992	0.999	0.510

Table S3: WW domain (PF00397) ρ sweep. $K = 420$, $K_{\text{des}} = 69$.

ρ	f_{eff}	f_{obs}	$\bar{a}_{\text{des}}$	D
1	0.164	0.131	0.152	0.690
5	0.496	0.184	0.487	0.689
10	0.663	0.226	0.661	0.676
50	0.901	0.277	0.901	0.657
100	0.947	0.303	0.947	0.649
200	0.973	0.327	0.974	0.643
500	0.990	0.360	0.991	0.642
1000	0.995	0.371	0.995	0.642

Table S2: β sweep at fixed ρ on Kunitz

Phenotype fraction and diversity as a function of β/β^* for three fixed ρ values on the Kunitz domain. f_{obs} and D are averages over 930 generated sequences. All conditions produced 100% valid residues.

S2 Supplementary Figures

Binding-loop per-position amino acid frequency heatmaps for the ω-conotoxin experiment are shown in Fig. 6 of the main text (positions 9–17 of the alignment). Additional SI figures for the cross-family PCA separation analysis and full entropy-curve families are described below.

Table S4: β sweep on Kunitz domain ($K = 99$, $K_{\text{des}} = 32$).

ρ	β/β^*	f_{obs}	D
10	0.50	0.471	0.581
10	1.00	0.539	0.547
10	1.50	0.573	0.525
10	2.00	0.597	0.508
10	3.00	0.613	0.484
50	0.50	0.506	0.573
50	1.00	0.582	0.537
50	1.50	0.631	0.514
50	2.00	0.653	0.499
50	3.00	0.700	0.469
200	0.50	0.535	0.564
200	1.00	0.613	0.527
200	1.50	0.648	0.503
200	2.00	0.677	0.488
200	3.00	0.706	0.463

PCA separation histograms. For each family, designated and background sequences were projected onto their top two principal components (from the full-family PCA). The separation index S was computed as the ratio of the inter-class distance to the intra-class spread (see Methods). Kunitz ($S = 0.20$) shows partial overlap; SH3 ($S = 0.34$) shows clear separation; WW ($S = 0.11$) shows extensive overlap, consistent with the large calibration gap.

pLDDT versus TM-score correlation. Figure S1 shows the joint distribution of ESMFold pLDDT and TM-score for all 250 predicted structures. SA-generated sequences cluster tightly in the high-pLDDT, high-TM-score region (upper right), while HMM-emitted sequences scatter broadly across lower values. The two metrics are positively correlated, confirming that ESMFold confidence is a reliable proxy for structural accuracy on this family.

Entropy-curve families across ρ. Figure 3 in the main text shows entropy curves for five representative ρ values on the Kunitz family. The full sweep ($\rho \in \{1, 2, 5, 10, 20, 50, 100, 200, 500, 1000\}$) produces a monotonically rightward-shifting family of sigmoid curves; the inflection point $\beta^*(\rho)$ increases from 3.85 ($\rho = 1$) to 10.22 ($\rho = 1000$), consistent with $\beta^* \propto \log K_{\text{eff}}$ predicted by the mean-field argument in Section 4.1.

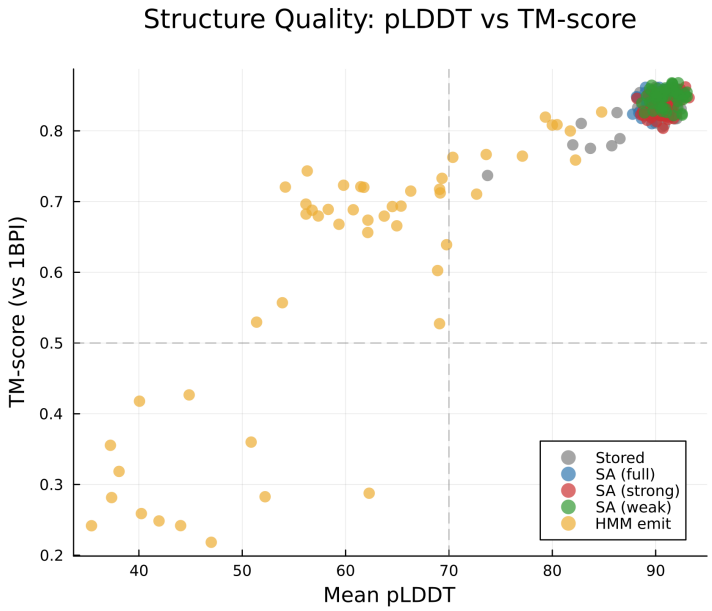

Figure S1: **pLDDT and TM-score are positively correlated; SA sequences cluster in the high-quality region.** Each point represents one predicted structure (50 per source). Stored (gray) and SA-generated (blue, red, green) sequences cluster tightly above pLDDT > 85 and TM-score > 0.8 (upper right quadrant). HMM sequences (orange) scatter broadly, with many falling below the pLDDT > 70 confidence threshold (vertical dashed line) and the TM-score > 0.5 same-fold threshold (horizontal dashed line).